



Brainstorming & its variations



Brainstorming?

- Brainstorming is a technique intended to rapidly generate a large number of solution ideas to a problem
- The method was first popularized in the late 1930s by Alex Osborn in a book called "Applied Imagination"

Structure of a Classical Brainstorming Session

- Ice-breaking (get to know each other)
- Phase I: Idea Purge Phase
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 - 4~12 members (diversity of thinking)
- Phase II: Idea Trigger Phase
 - Leader calls each member for their ideas
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 - If some ideas trigger other ideas, members should write them down
 - Repeat until no more new ideas are generated
 - Sort and organize the ideas

Creative
thinking

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- Phase III: Idea Evaluation & Solution Recommendation Phase
 - In its final stage of a brainstorming session, the group (or a different group) evaluate the ideas and select one or a few as the solutions to the problem proposed to the group

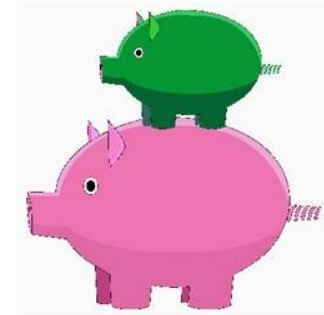
Judgmental
thinking

Two fundamental principles

- Delayed judgment:
 - Judgmental thinking vs. creative thinking
 - Judgmental thinking inhibits creative thinking
 - Delaying judgment is for generating as many ideas as possible
 - Delaying judgment eliminates fear (a mental barrier to creativity)
- Focus on quantity:
 - The “1/300” law (Makoto, 2007)
 - “When you generate 1000 ideas, only 3 of them are useful.”
 - Linus Pauling:
 - “One must endeavor to come up with *many* ideas — then discard the useless ones”

Basic rules for brainstorming

- Basic rules
 - No criticism: think first-judge later
 - Focus on quantity (even fragmented ideas or keywords are ok)
 - "Quantity bears quality"
 - Keep short & snappy
 - Don't settle for the first idea (avoid any fixation)
 - Support each other (never criticize)
 - Accept unusual ideas
 - Combine, improve, expand (hitchhiking/piggybacking)
 - Relax
 - Have fun
 - Laugh
 - The wilder, the better
 - Try to be creative (consciously try to **eliminate mental barriers**)
 - Reduce the social inhibitions that occur in groups, and therefore, stimulate the generation of new ideas
 - Try to create a dynamic synergy that will increase the creativity of the group

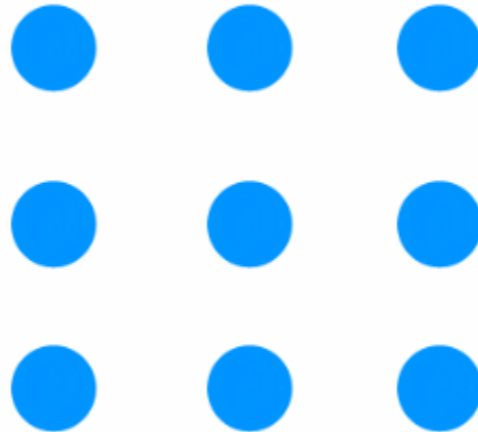


Types of Mental Barriers

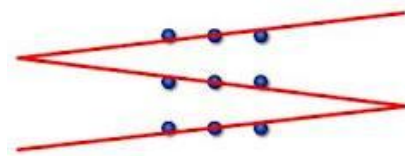
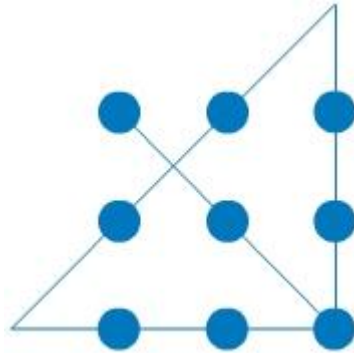
- False assumptions and nonexistent limitations
- Rigid or inappropriate solution strategies
- Overcomplicating problems
- Information overload or scarcity
- Miscommunication and misunderstanding
- Irrelevant or distracting associations (e.g., getting misled by false analogies)
- Emotional, cultural, and environmental barriers
- Fear
- Difficulty balancing divergent (creative) and convergent (critical) thinking
- Falling in love with an idea
- Paralysis from overabundance of resources

Nine circles problem

- Without lifting your pen, draw no more than 4 straight lines to cross all the nine circles



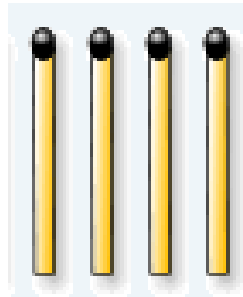
Some solutions



- Can you come up with other solutions?

4 matchsticks problem

- Using 4 identical matchsticks, can you create an image of two equilateral triangles?



Killer phrases that easily kill a brainstorming session

Beware of the following phrases: They stifle creativity and halt innovation!

- ~ It won't work!
- ~ We haven't the time.
- ~ We haven't the manpower.
- ~ We tried that before.
- ~ We've gotten along without it so far.
- ~ Do it the same way.
- ~ We have to be cautious.
- ~ Easy does it!
- ~ Damn good idea, but...
- ~ Let me sleep on it.
- ~ Why rock the boat?
- ~ That's risky thinking.
- ~ I've had five years of experience.
- ~ How could you be so stupid as to think....
- ~ Go ahead but I don't want to get mixed up with it.
- ~ We've never done it that way before.
- ~ Let's discuss it at some other time.
- ~ You didn't understand our problem.
- ~ We're too small for that.
- ~ Here we go again!
- ~ Let's think it over for awhile, and watch developments.
- ~ It's not our responsibility.
- ~ We never used that approach before.
- ~ Let's put it in writing.
- ~ Yes, but...
- ~ We have too many projects now.
- ~ Beware of your imagination

WHICH KILLER PHRASES DO YOU USE?

IDEA KILLER

B I N G O

But...	We've already tried that before.	It'll never fly.	Let me play devil's advocate here...	Let's not go off on a tangent.
You're setting yourself up for failure.	Sure it will...	In THIS economy?	Do you think we're made of money?	That's not a high priority right now.
Have you really thought about the implications?	That won't work because...		The only problem with that is...	Run an ROI, and get back to us.
Is this in line with our strategy?	The front line will never go for it.	You're kidding... right?	Yes, but...	Does anyone really care about that?
What you are really saying...	If it ain't broke...	Sure, in theory... but you don't think it'll really work	But how much is this idea worth?	Do we really have the resources for this?

Pause Before You Pounce.
Say "Yes" To An Idea First... See Where It Goes, Then Evaluate.

Idea-Sandbox.com

Blamestorming



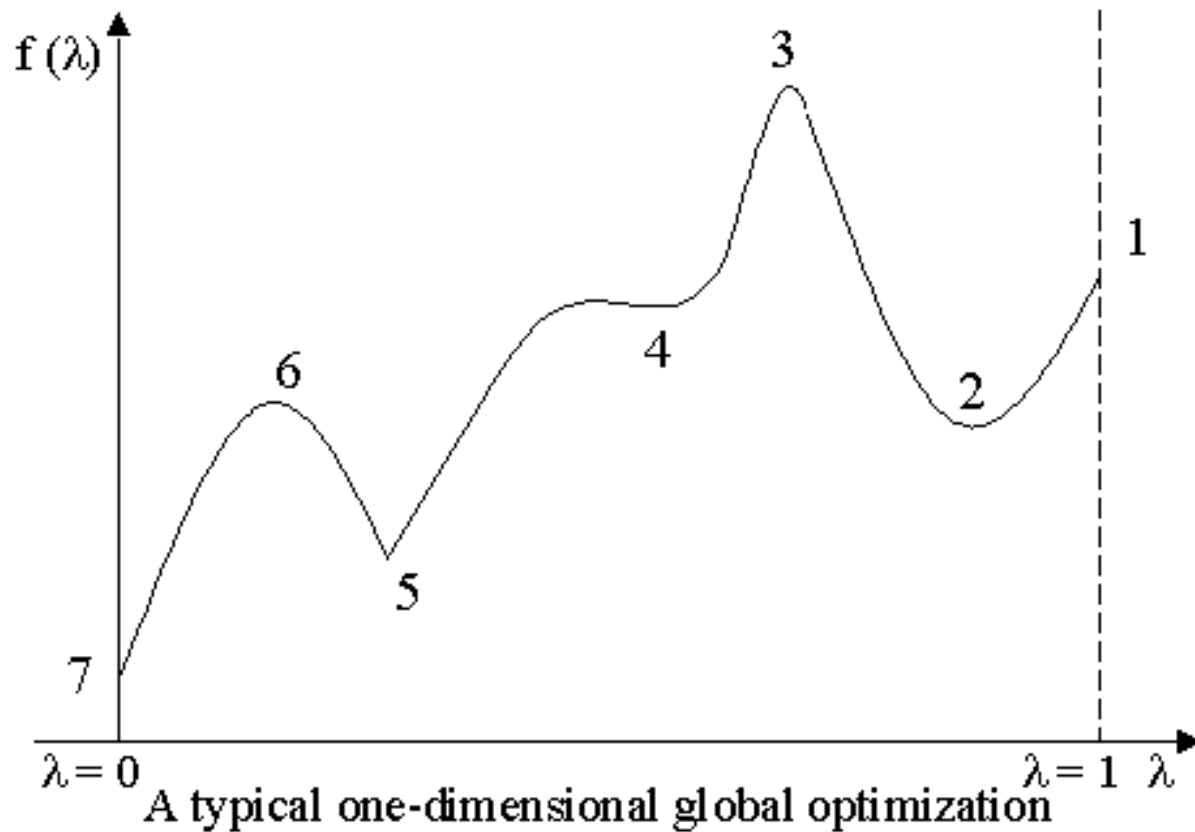
About the leader

- The leader's role is to record and write down all the ideas
- The leader should not direct or influence the brainstorming
- The leader is a facilitator or recorder, not a boss

Class Activity #1

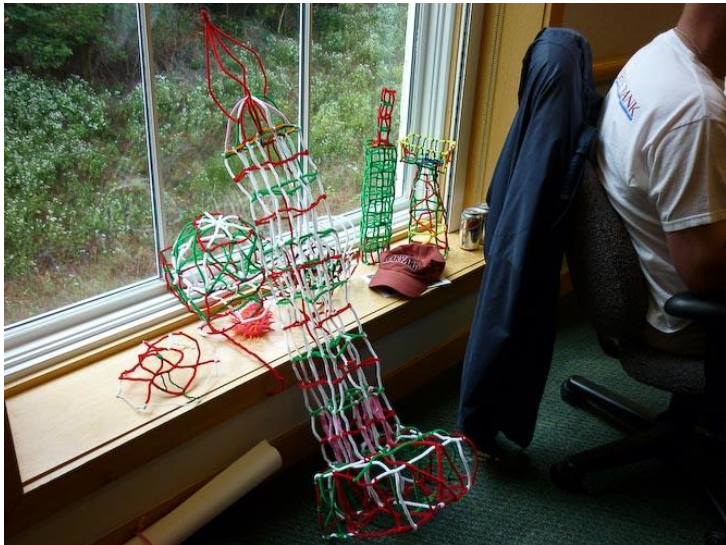
- Brainstorming exercise

An “optimization” analogy to brainstorming



Visualization and prototyping can be helpful

- In many cases, sketching ideas or building quick models can be more effective than simply writing them down.
- Some expert brainstormers even bring a “play kit” of Lego parts, play dough, duct tape, styrofoam, and other materials to quickly prototype their ideas during the session.



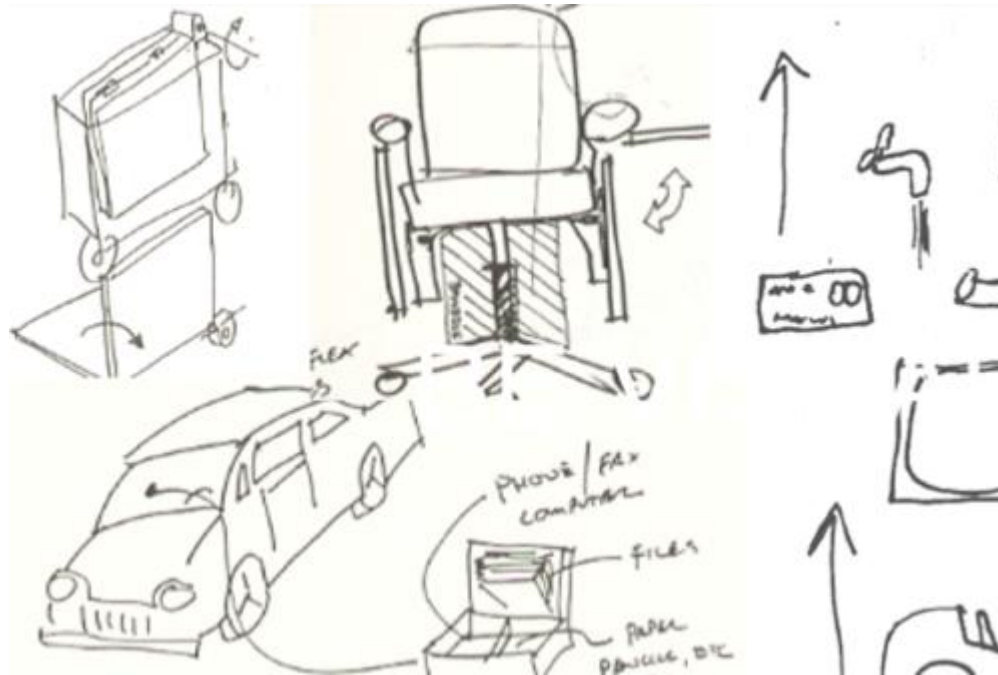
Play-doh



{Been There. Learned That}

Visual brainstorming (brain drawing)

- A variation of brainstorming
- Participants express their ideas visually instead of with words



Brainstorming and human information processing

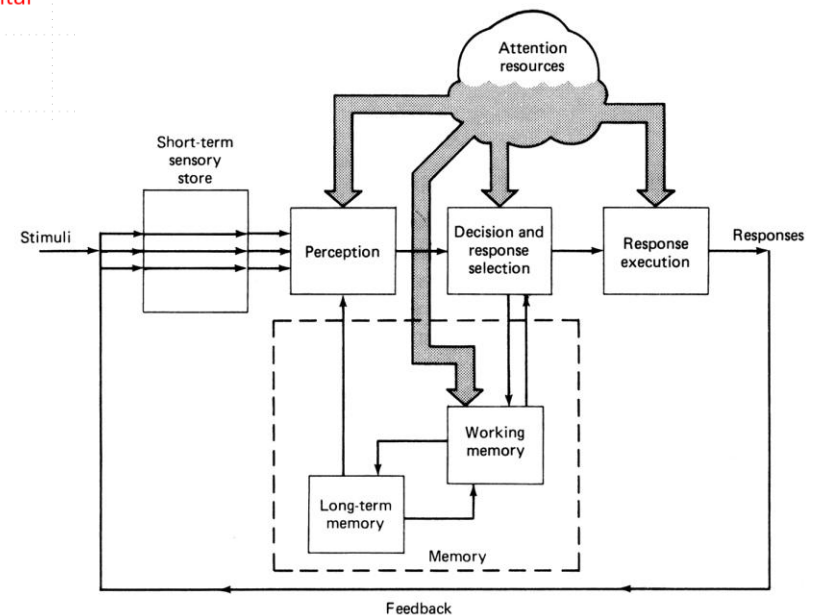
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Brainstorming and human information processing

- Ice-breaking and evaluation apprehension
 - Ice-breaking facilitates information sharing among group members
 - Reduces evaluation apprehension by creating psychological safety
 - Evaluation apprehension:
 - Reduces idea variety
 - Requires mentally demanding simulation of others' reactions
 - Tolls working memory (WM), consuming cognitive resources

Evaluation apprehension refers to the **fear or concern about how others will judge one's ideas, thoughts, or behavior** in a social or group setting.

Brainstorming and human information processing

- Sequential phases and cognitive efficiency
 - Sequential structure:
 - Idea purge phase → Idea trigger phase → Evaluation and recommendation.
 - Benefits:
 - Reduces mode switching between mental tasks
 - Minimizing mode switching supports the same goal as reducing the number of active channels during selective attention
 - Prevents cognitive overload from multitasking (e.g., generating + evaluating)
 - Helps group focus on one cognitive process at a time

Brainstorming and human information processing

- Idea purging and long-term memory (LTM)
 - Idea purging = Retrieving information from LTM
 - Influenced by:
 - How the problem is defined
 - Perceptual style and individual differences
 - Bottom-up and top-down processing
 - Internal knowledge networks
 - Decision making involved in selecting relevant ideas from internal stimuli

Brainstorming and human information processing

- Writing aids cognition
 - Writing down ideas:
 - Offloads WM → Frees capacity for manipulation.
 - Facilitates embodied cognition:
 - Spatial proximity of written ideas → Natural associations

Brainstorming and human information processing

- Idea triggering and WM
 - Triggering relies on WM for mental manipulation
 - Selective attention when choosing ideas on shared screen/board
 - Influences:
 - Bottom-up (visual salience)
 - Top-down (goals and expectations)
 - Fixation = Failure to move attention away from limited cues.
 - Focused attention needed to manipulate selected items

Brainstorming and human information processing

- Production blocking and turn-taking
- Participants must wait to speak:
 - Bottlenecks idea flow
 - Creates production blocking
- Waiting period:
 - Increases risk of forgetting ideas (WM loss)
 - Multitasking required: Listen + hold idea in WM

Brainstorming and human information processing

- Evaluation phase and bias
 - Idea evaluation and solution recommendation = Decision making phase.
 - Prone to bias:
 - Social dynamics (dominance, conformity)
 - Premature filtering of ideas.

Brainstorming and human information processing

- Summary
 - Group brainstorming involves complex cognitive activities:
 - Retrieval from LTM
 - WM manipulation
 - Decision making
 - Attention control
 - Cognitive bottlenecks:
 - Evaluation apprehension
 - Production blocking
 - Multitasking demands
 - Structuring and writing can help reduce cognitive load and improve output

Strengths of brainstorming

- Generates a large number of ideas in a short time
- Encourages free thinking and creativity
- Builds on each other's ideas (synergy and idea cross-pollination)
- Creates a supportive group atmosphere that meets social-emotional needs, fosters motivation, and strengthens cohesion (if facilitated well)
- Reduces fear of judgment when the "no criticism" rule is followed
- Provides diverse perspectives that individuals alone may not generate

Weaknesses of brainstorming

- May be limited to relatively simple or well-defined problems
- Generated ideas can be low-quality or superficial without proper facilitation
- May be less efficient than individual work for certain tasks (low return on investment of time and effort)
- Risk of group dynamics issues:
 - Some members may dominate the discussion
 - Others may feel shy, intimidated, or reluctant to contribute
 - Conflicts can arise when volatile issues are discussed
- Delaying judgment is difficult in practice, leading to premature evaluation of ideas
- Hard to give due credit to individuals for contributions
- Typically generates only one idea at a time, slowing the process
- Susceptible to **groupthink**, where conformity pressure reduces originality
- Can encourage quantity over quality, with too much focus on idea counts
- Requires skilled facilitation; without it, sessions often lose focus or drift

Variations of brainstorming

Individual Brainstorming

- Brainstorming can also be done by a single individual.
- Recent studies suggest that when multiple people brainstorm individually and then pool their ideas, the results can sometimes be better than those from a group brainstorming session.

Brainwriting (Method 635)

- A variation of brainstorming designed to overcome its weaknesses

BRAINWRITING



Brainwriting (Method 635)

▷ 6-3-5 Method

1. 6-3-5 means: 6 in group/3 ideas per round/5 minutes per round.
2. Divide everyone into groups of about 6. Too many in a group is unmanageable, too few restricts the generation of ideas.
3. Each participant starts with a prewritten brainwriting form. The problem to be addressed is written at the top of the form (see example).
4. In the first round, participants have 5 minutes to write 3 ideas in the top boxes (1 per box) of the brainwriting form. Often the problem is known ahead of time and the participants come in with the 3 ideas already developed. If this is the case, this initial time can be shorter.
5. At the end of each round, the form is passed to the person on the right. As each person gets a form from the person on the left, they read all the ideas on the sheet and then add three new ones. The new ideas can be completely new, or can be variations of ideas already on the sheet, or can be additional developments to ideas already on the sheet. Ideas from other participants should foster new ideas. There is no talking or discussion during these rounds.
6. The process is completed when each participant gets his own form back, now filled up with many ideas.
7. The last step is to sort the ideas using a clustering method

Brainwriting form

Problem statement: How to...			
	Idea 1	Idea 2	Idea 3
1			
2			
3			
4			
5			
6			

Brainwriting (Pool Method)

Pool Method

1. Each participant gets a form. Problem is written on form.
2. 5 – 8 in group.
3. Each person writes three ideas at top and puts sheet in center of table.
4. Participants take new sheet out of center pile and add to it.
5. No rounds. Put sheets back and get new sheets at own pace.
6. Process completed at end of pre-determined time (e.g. 30 min).
7. Sort ideas.

Comparison of group brainstorming, individual brainstorming and brainwriting

Method	Advantages	Disadvantages
Group Brainstorming	- Real-time idea exchange and dynamic interaction - Builds on group energy - Can spark spontaneous ideas	- Production blocking: Since only one person can speak at a time, others must wait, which can slow down idea generation and lead to information loss in working memory as participants suppress or forget ideas while waiting - Evaluation apprehension—the fear of being judged—requires mental simulation and decision-making, which is cognitively demanding and can hinder idea generation - Requires multitasking: listening while thinking, which can overload the cognitive system due to overlapping cognitive resources
Individual Brainstorming	- Time-efficient (no waiting) - No evaluation apprehension - Full focus on one's own thoughts	- No stimulation from others' ideas - No immediate feedback or collaborative idea building
Brainwriting	- Everyone writes ideas simultaneously (parallel process) - No evaluation apprehension - Encourages structured building on others' ideas - Promotes equal participation	- Less spontaneous and dynamic than spoken brainstorming - Can be slower in large groups (ideas take time to circulate) - Less verbal interaction

Sorting and organizing ideas (brainstorming & brainwriting)

↳ Cluster Technique

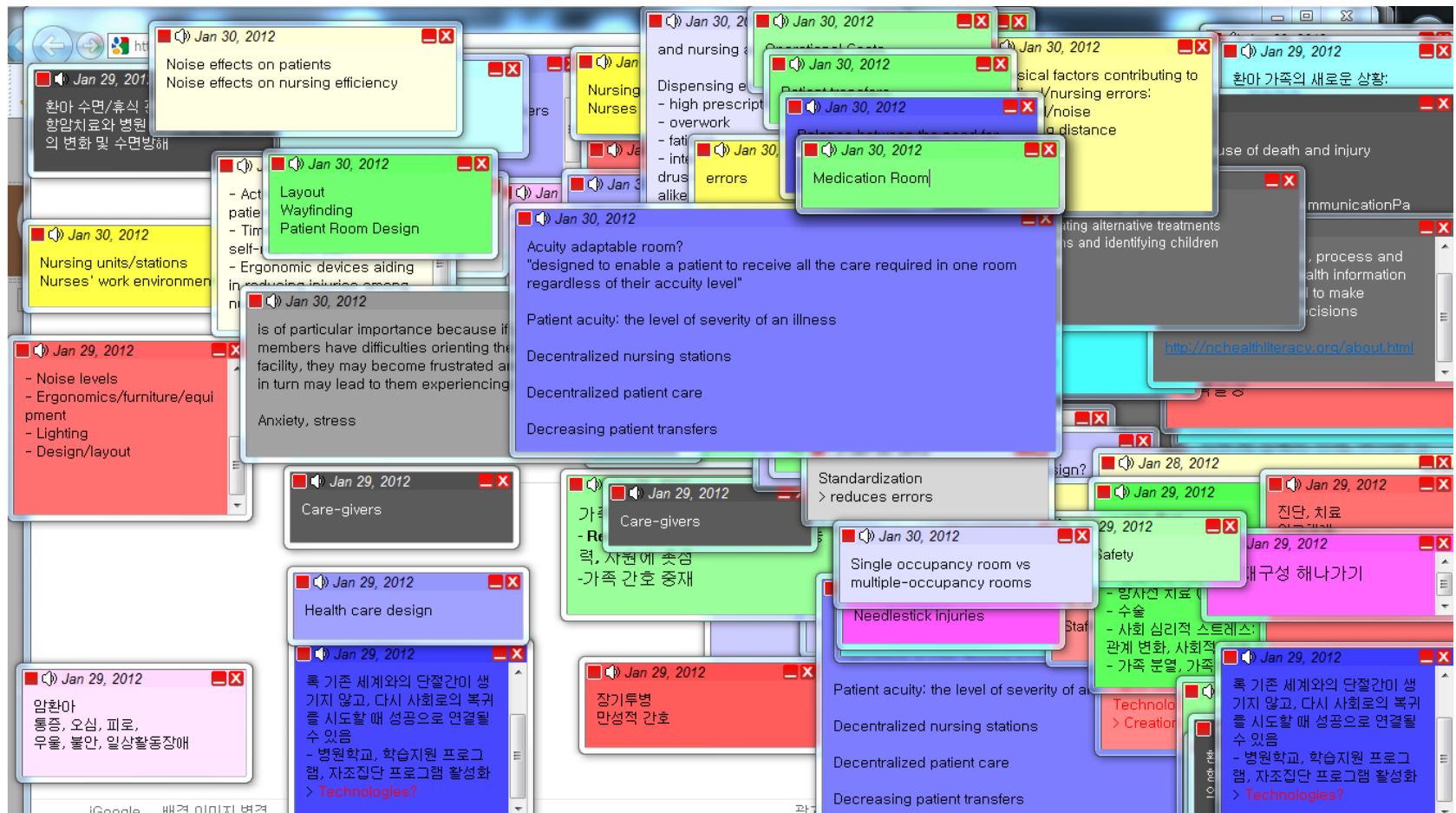
1. Immediately after the brainwriting session, all ideas are copied onto Post-Its.
2. Participants, as a group or individually, begin arranging ideas into “clusters.” This is simply a sorting process; there should be no discussion of which ideas are better or worse. Some require absolute silence at this point.
3. The group agrees on (or an individual assigns) a label to put on each cluster.
4. If an idea can fit into two or more clusters, duplicates may be made.
5. Clusters can stand individually if the goal is just to get a bunch of ideas. If some consensus must be reached, discussion can begin about which clusters to merge or eliminate.

Sorting & organizing ideas



Virtual Post-It

- Sticky Notes



Reverse Brainstorming

Reverse brainstorming is a way to bring a new perspective to idea generation that can be especially helpful when a team is cynical about a project or lacking energy. If you have members of your team who are hostile or worn out, this approach can take advantage of the hostility or fatigue. The process is simple. Let's say that your senior managers want to increase customer loyalty for the new e-commerce site that has just been launched. In traditional brainstorming, you would ask the question "What could we do to improve customer loyalty for our new site?" In reverse brainstorming, you can ask the negative question "What could we do that would reduce customer loyalty for our site?" or perhaps more strongly "What could we do to drive customers away from our site?" This negative question might yield answers like:

1. Require users to read every line of the software license.
2. Make it hard to log in.
3. Ask for lots of personal information.
4. Use deceptive defaults to make more money for us (for example, default to a 2-year subscription and put the default where it is likely to go unnoticed).
5. Hide the fact that products might be out of stock.
6. Use deceptive out-in/opt-out labels.
7. Hide things behind general links; e.g., putting opt-out features behind a "More info..." link.
8. Use double negative labels to trick users.
9. Not be clear what links go to ads and what links go to actual content.
10. Add things to the shopping cart in a sneaky way.



CYNICISM

He learned it from watching YOU.

motifake.com

Reverse Brainstorming

When to Use:

This method can be used simply to provide a different perspective on a particular problem. It is also useful when you have a team with low morale, serious cynicism, or open hostility. Reverse brainstorming can sometimes bring smiles to a group that has struggled with new ideas to old and difficult problems. You can use this method as a complement to other ideation methods.



Reverse Brainstorming

Procedure:

1. Develop a question for brainstorming and then convert it to a negative question.
2. Let the brainstorming team know about the specific question and ask them to do a little homework and generate a few items before the group session.
3. Write the questions on the board or display it on the online brainstorming tool.
4. Describe the guidelines for the brainstorming session.
 - a. No criticism of negative ideas.
 - b. New and wild negative ideas are welcome.
 - c. No war stories; just describe the negative item so that everyone understands what you mean (war stories waste time and prevent others from expressing their ideas – this is called “production blocking” in the brainstorming literature)
 - d. Only one person can speak at a time.
5. Conduct a short warm-up exercise using a negative question that is not related to your main question.
6. Ask if there are any questions.
7. Begin the reverse brainstorming.
8. After you generate the negative ideas, schedule a “conversion” meeting where you convert the negative ideas into positive ideas as shown below. Each negative idea might yield more than one positive idea.

Reverse Brainstorming

Reverse Brainstorming Ideas	Converted Ideas
Require users to read every line of the software license.	Simplify the license and use plain English
Make it hard to log in.	Eliminate log-ins; if the person is an intermittent user, provide an easy way to get at passwords and even user names; make the username email.
Ask for lots of personal information.	Minimize the amount of personal information required at all stages of purchasing; Explain why certain information is needed ("Why we need a phone number?")
Use deceptive defaults to make more money for us (for example, default to a 2-year subscription and put the default where it is likely to go unnoticed).	Use good defaults that do not appear to take advantage of users; be clear about what the defaults are; develop a list of best practices for choosing defaults
Hide the fact that products might be out of stock.	Provide a warning about out-of-stock; provide an in-stock warning; provide a link to the closest alternative

Reverse Brainstorming Exercise

- Refrigerator design

Reverse brainstorming

- Reverse brainstorming helps overcome fixation
- Fixation = Getting stuck on a narrow set of ideas due to focused attention
- Reversing the problem changes how it is viewed, which changes what is retrieved from long-term memory
- Ideas generated from the reversed problem are still related to the original one
- This approach helps explore new areas of the solution space and break free from fixation

Bodystorming

- Bodystorming is a technique of physically experiencing a situation to generate new ideas. It involves setting up an experience — complete with necessary artifacts and people — and then physically “testing” it.
- It can also include changing your physical space during ideation. The focus is on how you interact with your environment and the choices you make while engaging in it.

Bodystorming

How to bodystorm?

1. Get up
2. Do it

It's pretty simple, really. Get physical! If you are trying to ideate in the context of hospital patients, try walking through the experience to come up with new ideas. If you are designing products for the elderly, rub some Vaseline on your glasses to view the world through older eyes. Bodystorm by moving around and becoming aware of the physical spaces and experiences related to your solutions. Pay close attention to decision-making directly related to your environment and related emotional reactions. Dig into the "WHY"!

Bodystorming tools

Ford's 3rd age suit



Bodystorming tools

Experiential design with VR

